armfield

C11-MkII Pipe Networks

The Pipe Networks is a free-standing accessory to the F1-10 Hydraulics Bench.

Demonstrates the characteristics of flow through different arrangements of pipes and the effect of changes in pipe diameter on the flow through a particular network.

PRODUCT CODE: **C11-MKII**

Categories: C Series - Advanced Fluid Mechanics, C11 MkII - Pipe Networks, Educational

Tags: Applied Fluid Mechanics , C Series , Characteristics Of Flow , Electronic Pressure Meter , F1-10 Hydraulics Bench , Head Loss , Parallel , Series

DESCRIPTION

The **Armfield** Pipe Networks is a free-standing accessory to the F1-10 Hydraulics Bench which demonstrates the characteristics of flow through different arrangements of pipes and the effect of changes in pipe diameter on the flow through a particular network.

The permanent arrangement of PVC pipes and fittings is mounted on a free-standing support frame designed to stand alongside an F1-10 Hydraulics Bench in use. Connection to the F1-10 is via a reinforced flexible tube and threaded union with ‘O’ ring seal, enabling connection to the F1-10 without the use of tools.

Isolating valves enable a wide range of different series, parallel and mixed pipe configurations to be created without draining the system. Flow into the network and flow out from the network at each outlet can be individually varied to change the characteristics of the system.

All clear acrylic test pipes are installed using threaded unions with ‘O’ ring seals that allow the pipes with different diameters to be repositioned without the use of tools.

Self-sealing quick-release fittings at strategic points in the network allow rapid connection of the digital hand-held pressure meter, allowing appropriate differential pressures to be measured.

Flow leaving any of the outlets in the network is measured using the volumetric facility incorporated on the F1-10 Hydraulics Bench.

TECHNICAL SPECIFICATIONS

Length of pipes: 0.70m

Inside diameter of pipes:

14mm (1x)
10mm (1x)

9mm (2x)
6mm (1x)

Nominal inside diameter of manifold: 22.4mm

Differential pressure measurement: Digital pressure meter (alternative units selectable)

- **Range:** 0 – 2000cmH_2O (0 – 2000 mBar)
- **Resolution:** 1cm H_2O (1 mBar)

FEATURES & BENEFITS

- Measurement of head loss versus discharge for different sizes of pipes
- Characteristics of flow through interconnected pipes of different sizes
- Characteristics of flow through parallel pipe networks
- Characteristics of flow through series pipe networks
- Application of doubling pipes on existing networks to increase flow rate
- Characteristics of flow around a ring main and the effect of changes in supplies and off-takes

Experimental contents

- Measurement of head loss versus discharge for different sizes of pipes
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Video (English y Español)

Complementary products

- **C1-MKII:** Compressible Flow Bench
- **C3-MKII:** Multi-pump Test Rig
- **C4-MKII:** Multi-purpose Flume
- **C6-MKII:** Fluid Friction Measurements
- **C7-MKII:** Pipe Surge and Water Hammer
- **C9:** Flow Meter Demonstration Unit
- **C10:** Laminar Flow Table
- **F1-18:** Energy Losses in Pipes
- **F1-21:** Flow Meter Demonstration
- **F1-22:** Energy Losses in Bends and Fittings

Ordering specification

- Specifically designed to allow the setting up of a wide range of different pipe arrays (networks)
- Pipe network mounted on free-standing support frame for use alongside an F1-10 Hydraulics Bench
- Clear acrylic test pipes are all 0.70m long with inside diameters of 1x 6mm, 2x 9mm, 1x 10mm, 1x 14mm
- Includes hand-held electronic pressure meter with self-sealing quick-release connections to the pipe network
- Flows into and out from the appropriate network can be varied individually

Requirements

Electrical supply: See Hydraulics Bench

The F1-10 requires an electrical supply,
please refer to the F1 data sheet for details.

- F1-10-A: 220-240V / 1ph / 50Hz / 10A
- F1-10-B: 110-120V / 1ph / 60Hz / 10A
- F1-10-G: 220V / 1ph / 60Hz / 10A
- F1-10-2-A: 220-240V / 1ph / 50Hz / 10A

- F1-10-2-B: 110-120V / 1ph / 60Hz / 10A
- F1-10-2-G: 220V / 1ph / 60Hz / 10A

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 1.4m³

Gross Weight: 240Kg

Dimensions

Length: 0.785m

Width: 0.656m

Height: 1.380m

Order Codes

C11-MKII

C11-MKII-10

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